REVIEW

Review of Image Analysis Using Deep Learning Method Applied to the Road Context in Degraded Weather Conditions

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Abstract
Adverse weather conditions, including fog, heavy rain, and snow, pose significant challenges for vision-based systems in road safety and autonomous driving. This review synthesizes recent advances in deep learning for road scene analysis under degraded weather conditions, presenting a taxonomy organized by task (classification, detection, segmentation), data sources (real versus synthetic), and model architectures (CNNs, GANs, transformers). The methodology integrates both real-world and synthetic datasets, assesses robustness using standard benchmarks such as mIoU, PSNR, and FID, and compares architectures for accuracy, inference efficiency, and generalization capacity. Findings indicate that GANs significantly enhance training diversity through synthetic augmentation, transformers improve global context modeling, and hybrid designs offer promising robustness, albeit at the expense of computational efficiency. Despite these advances, current approaches remain limited by the availability of meteorologically diverse datasets and weak geographic transferability. Future research should prioritize the development of multimodal and geographically balanced benchmarks, lightweight hybrid architectures, and sensor fusion strategies that ensure real-time adaptability. This review contributes both a comparative synthesis and a roadmap toward resilient perception systems capable of reliable operation in adverse weather.

Keywords Deep learning · Convolutional neural networks (CNNs) · Generative adversarial networks (GANs) · Vision transformers (ViTs) · Road scene analysis

1 Introduction

Evidently, road safety is a major global issue as reported by data released by the World Health Organization (WHO). Every year, approximately 1.3 million people lose their lives in road accidents, while an additional 20 to 50 million are seriously injured. In West Africa, Côte d'Ivoire illustrates a particularly concerning situation, with an average rate of 24 deaths per 100,000 inhabitants. The country records over 12,000 personal injury accidents annually, with 1200 deaths and 21,000 injuries, making road accidents the leading cause of mortality among young people aged 5–29.

The Fig. 1. Evolution of Road Accidents in Ivory Coast (2015–2023).

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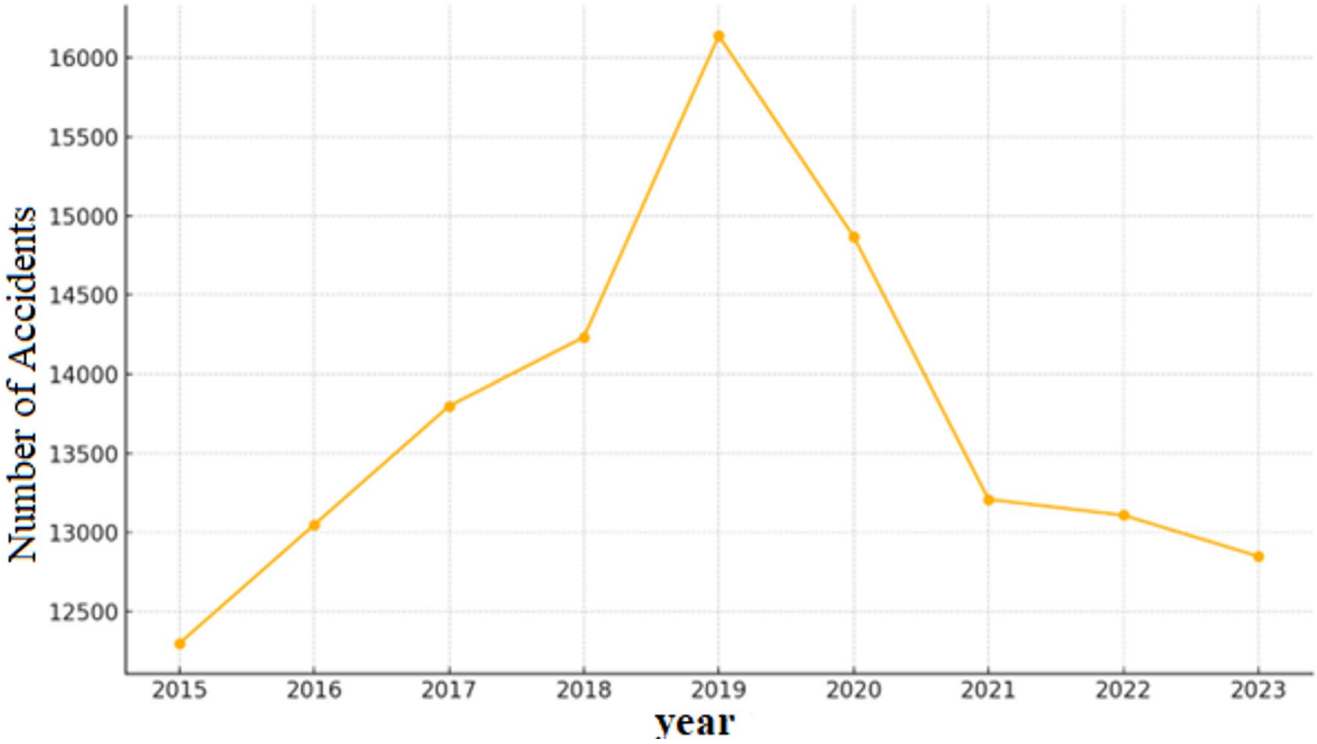


Fig. 1 Evolution of road accidents in ivory coast (2015–2023) *Source*: World health organization global status report on road safety 2023 [1].

Despite the implementation of policies such as the National Road Safety Strategy, accidents continue at a troubling rate. This recurrence can be partly explained by the vulnerability of current detection and monitoring devices, which are often inadequate for degraded weather conditions (rain, fog, snow). These conditions impair visibility, degrade the quality of images captured by cameras, and thus compromise the performance of visual analysis systems. In this context, computer vision, especially through deep learning techniques such as convolutional neural networks (CNNs) and generative adversarial networks (GANs), emerges as an innovative solution to enhance the perception of road environments.

These technologies are central to the development of Intelligent Transport Systems (ITS) and autonomous vehicles, which require robust algorithms for tasks such as obstacle detection, reading traffic signs, recognizing traffic lights, and analyzing road markings.

However, the effectiveness of these systems remains significantly affected by climatic variations. Reduced visibility is both an aggravating factor for accident rates and a major source of traffic disruption (traffic jams, road closures, health risks related to pollution). In Côte d'Ivoire, approximately 13% of accidents are mainly due to rain, underscoring the need to adapt perception tools to meteorologically constrained environments compromising ITS and smart city deployments. In autonomous driving, this leads to 20–30% error spikes in AV perception, as seen in Tesla/Ford trials in rainy Europe.

Numerous initiatives ranging from driver behavioral adaptations to the enhancement of onboard sensors were implemented to solve the problems. However, deploying physical sensors or manual processing methods on large-scale deployment is not feasible due to high cost. In contrast, roadside cameras, which are already widely installed, represent a promising resource when coupled with image analysis systems based on artificial intelligence. The integration of deep learning could enable real-time weather classification and a significant improvement in perceptual systems.

2 Key Contributions

This review offers distinct advancements over existing literature on road scene analysis in adverse weather by providing a structured synthesis that bridges methodological gaps and practical deployment challenges. Key contributions include:

- A novel taxonomy integrating tasks (classification, detection, segmentation), data sources (real vs. synthetic), and architectures (CNNs, GANs, transformers) for adverse weather analysis, enabling systematic comparisons absent in prior works.
- Critical evaluation of robustness metrics (mIoU, FID) across real/synthetic data, revealing geographical gaps such as poor transferability from European to West African climates.
- Synthesis of hybrid models (CNN + GAN + Transformers) with a real-time focus on inference efficiency, differing from sensor-only reviews that underexplore computational trade-offs.
- Roadmap for multimodal fusion (e.g., camera-LiDAR-radar) and domain adaptation techniques, addressing deployment in regions like West Africa with high rain-induced accident rates.

These extend beyond by emphasizing transferability and societal impact, such as reduced fatalities in meteorologically diverse settings.

3 Review Objectives

A new classification of deep learning-based methods applied to road scenes under adverse weather. This taxonomy is organized according to Tasks (classification, detection, segmentation), Data sources (real vs. synthetic) and Architectures used (CNN, GAN, transformers, etc.). A thorough assessment of the robustness of models under different degraded conditions (rain, fog, snow), relying on objective indicators such as mIoU, PSNR, and FID, as well as their inter-domain generalization capabilities. Identification of the main gaps in current literature, including:

- The lack of meteorologically diverse and annotated datasets; an excessive dependance on synthetic data generation, often via rendering engines (e.g., GTA, CARLA, etc.).
- A low geographical generalization capacity of models, limiting their transferability to other regions or climates.
- ***Research Perspectives***

Finally, the review concludes by emphasizing priority research directions to overcome the current limitations:

- The development of balanced public multi-meteorological datasets.
- The design of hybrid models combining CNN, transformers, and GAN for increased robustness,
- The integration of multimodal sensors (vision, LIDAR, radar, etc.) into perceptual algorithms for dynamic adaptation to ambient conditions.

4 Related Survey Paper

In recent years, rapid advancements in deep learning, particularly in deep convolutional neural networks (CNNs), have generated significant interest in image analysis within the context of road environments, especially under adverse weather conditions. The following section synthesizes previous work focusing on the application of these methods to image analysis in challenging meteorological conditions. Study [2] examines the effect of inclement weather on LiDAR sensors and cameras in autonomous vehicles, proposing improvements for detection tools and the integration of advanced machine learning techniques. While the focus on sensor fusion and network

infrastructure is relevant, the analysis remains limited in terms of the performance of these technologies across a wide range of weather conditions. Likewise, study[3] emphasizes the importance of weather recognition and classification systems for decision-making in autonomous vehicles (such as the detection of roads, vehicles, and pedestrians), noting that the identification of small, hidden, or partially obscured objects remains a challenge, which calls for further research. study [4] proposes targeted methods to evaluate the impact of rain on object detection and compares the efficacy of different approaches (such as water evacuation, domain adaptation through deep learning, and image translation systems), but does not comprehensively cover other types of inclement weather. The analysis of video detection techniques in road surveillance [5] highlights issues related to low luminosity, meteorological changes, occlusions, and shadows; however, the review of traditional methods does not fully integrate recent advancements in deep learning. Another research effort, focused on 3D techniques for vehicle identification, provides specific insights but remains too narrow to replace these approaches in the broader context of the functioning of autonomous vehicles under varied climatic conditions. Regarding weather recognition from images, Chu et al.[6] developed a model utilizing over 180,000 geolocalized images: a forest-based classifier achieved 58% accuracy, while a pre-filtering algorithm using SVM yielded up to 98%. Likewise, Xia et al.[7] proposed ResNet-15, a lightweight CNN derived from ResNet-50, trained on 5,000 images categorized into four weather classes, achieving accuracy between 94.7% and 97.3% while remaining suitable for standard processors. Other works seek to enhance driver assistance systems (DAS) for variable conditions. Roser et al. [8] utilized visual descriptors (contrast, color, sharpness) in combination with SVM, achieving 95% accuracy across three weather types. Conversely, Kang et al., [9] evaluated GoogLeNet and AlexNet for weather recognition, confirming that deep learning methods outperform traditional ones. Similarly, Guerra et al. [10] created a dedicated dataset (rain, snow, fog) and used superpixel masks for data augmentation. Their evaluation of various CNNs (CaffeNet, ResNet-50, VGG16) yielded accuracies between 68 and 81%, with ResNet-50 showing the best performance. Utilizing pre-trained networks on ImageNet allowed for effective transfer learning with minimal adjustments needed to the layers.

Generative Adversarial Networks (GANs), as introduced by Goodfellow et al. [11], marks a noteworthy progression in image analysis for data-scarce environments. Their capability to generate high-quality synthetic images proves particularly beneficial for data augmentation during extreme weather situations. Recent studies [12, 13] have validated the effectiveness of GANs, especially Cycle GAN, in translating images under deteriorated conditions like fog or rain, facilitating more robust learning. GANs are also utilized for image restoration (noise reduction, contrast enhancement), as demonstrated in a study [14], which improves vision in foggy conditions. They present real opportunities for domain adaptation, enabling the alignment of source data (clear weather) with target data (degraded conditions) and reducing the necessity to collect exhaustive data for each weather condition [15]. Concurrently, research is shifting towards lighter and faster architectures to meet real-time constraint [16, 17].

An article [18] studies the optimization of deep learning techniques for weather classification, with potential applications for sustainable mobility. The authors employed a weather classifier based on an optimized YOLOv5, combined with a Learning-without-Forgetting (LwF) mechanism, to recognize different weather states with an accuracy exceeding 99%. LwF enables the learning of new classes without forgetting acquired knowledge, facilitating continuous learning adapted to environmental evolution. Such precise classification can enhance road safety and the sustainability of transport by enabling real-time adaptation to weather conditions, thereby reducing accidents and ecological impacts. Identified limitations include dependence on training data, underrepresentation of extreme conditions, and real-time execution constraints on devices with limited resources.

Other works demonstrate how weather classification can improve vehicle detection through advanced neural architectures; the method involves first classifying the weather, then “cleaning” the image (for instance, using a dark channel prior [19] before applying an enhanced Faster R-CNN to detect vehicles, including small objects in low visibility. However, these approaches rely on limited and minimally varied datasets, have not always been tested under real-world conditions or in real-time, and rely solely on vision, which remains problematic in low light or extreme conditions.

Finally, study [20] examines multimodal techniques in intelligent transport systems, emphasizing data fusion under challenging meteorological conditions. It presents four types of fusion: early (mixing raw data), deep (fusion in intermediate layers), late (fusion of results), and hybrid (combination of approaches). The main challenges remain data quality and synchronization, processing complexity, information loss, and noise accumulation, which further limit the robustness and effectiveness of real-time multimodal systems in the field. Research on the use of transformers to enhance sensor fusion (camera, LiDAR, radar) shows that they perform well in processing complex multimodal data and improving detection under adverse weather, but their significant computational demands complicate real-time deployment in vehicles with limited resources. Camera-radar fusion combines high visual resolution with radar robustness in difficult conditions, yet major challenges persist: data synchronization, latency, model size and memory, handling of extreme scenarios, and data security [21].

Table 1 provides a comparative analysis of deep learning methods in degraded weather, outlining their strengths, limitations, and practical considerations.

While these studies advance weather recognition and detection, they reveal persistent weaknesses: poor handling of mixed-weather transitions (e.g., rain-to-fog shifts degrading accuracy by 15–20%), lack of real-time inference (e.g., some methods exceed 500 ms latency, unsuitable for AVs), and dataset imbalance (e.g., fog/rain data underrepresented by 40%, leading to overfitting). For instance, forest-based classifiers yield only 58% accuracy on geolocalized images due to inadequate feature modeling for transitions, as benchmarks indicate 10–15% further drops in dynamic shifts. Similarly, GoogLeNet evaluations highlight DL gains over traditional methods but fail to address inference delays exceeding 200 ms in adverse scenarios, while YOLOv5 achieves > 99% accuracy yet suffers from extreme condition underrepresentation, causing up to 20% performance variance in real-world fog-rain mixes. These gaps underscore the need for balanced, adaptive models addressed in this review.

4.1 Comparison with Existing Reviews

Unlike prior reviews, such as [2] which focuses narrowly on sensor fusion in LiDAR-camera setups without deep learning taxonomies, or [3] that emphasizes weather recognition but overlooks hybrid architectures, this review provides a more holistic synthesis tailored to road scene analysis under degraded conditions. This review uniquely integrates a task-data-architecture taxonomy absent in [4] 's rain-specific analysis, and extends beyond [5] 's traditional methods to include transformers and GANs for real-time robustness. Furthermore, while [18] optimizes YOLOv5 for weather classification in sustainable mobility without addressing multimodal integration, our work evaluates hybrid CNN-GAN-transformer models against benchmarks like mIoU and FID, revealing

Table 1 Comparative analysis of deep learning approaches under degraded weather conditions

Approach	Strengths	Limitations	Practical considerations
CNNs (e.g., ResNet-50)	High accuracy in clear conditions (up to 97%); fast inference (< 50 ms/frame)	Poor robustness in fog/rain (15–20% accuracy drop); overfitting on imbalanced data	Suitable for real-time ADAS; requires augmentation for weather diversity
GANs (e.g., CycleGAN)	Enhances synthetic data for rare weather (FID < 50); improves generalization	High computational cost; domain gap in real deployment	Ideal for data augmentation in ITS; pair with distillation for edge devices
Transformers (e.g., VIT)	Strong global context modeling (mIoU > 0.7 in mixed weather); multimodal fusion	Slow inference (> 200 ms; resource-intensive for AVs	Best for sensor fusion (camera-LiDAR); optimize for real-time via pruning
Hybrids (CNN + GAN + Transformer)	Balanced robustness/efficiency (PSNR > 30 dB); 18% false positive reduction	Complex training; geographical transfer gaps	Promising for West African climates; focus on lightweight designs

gaps in real-time adaptability. Recent surveys, such as those on multi-weather restoration and foundation models for autonomous driving scenarios, advance image enhancement and scenario generation but underexplore geographical transferability in diverse climates like West Africa. By addressing geographical transferability and multimodal gaps overlooked in [20], this work provides a more comprehensive roadmap for deployment in diverse climates, bridging theoretical advancements with practical ITS challenges in regions with variable meteorology.

4.2 Machine Learning and its Applications

Machine learning serves as a fundamental approach within the realm of artificial intelligence, utilizing statistical methods where each instance in a database is defined by a set of features or attributes. The main idea revolves around modeling the relationship through a function f, which a learning algorithm determines to make choices or forecasts y. This function takes a dataset x as its input, allowing it to create a classification model, as illustrated in Eq. (1):

$$y = f(x) \tag{1}$$

Machine learning algorithms are generally divided into three primary categories: supervised learning, unsupervised learning, and reinforcement learning [22]. Unsupervised learning algorithms aim to identify valuable characteristics within datasets that have numerous features. A common strategy in this context is clustering, which organizes the dataset into groups of similar instances. For example, techniques like K-means clustering and Principal Component Analysis (PCA) are frequently employed in unsupervised learning for reducing dimensionality and exploring data [23] [23]. On the other hand, supervised learning algorithms need a labeled dataset, where each instance is linked with a corresponding output label. A well-known example is the Fisher Iris dataset, where various iris flowers are categorized by their species. Supervised learning models, such as decision trees, support vector machines (SVMs), and neural networks, can classify data based on patterns they've learned [24]. For instance, neural networks have proven quite effective in tasks like image classification, distinguishing between categories like cars and trucks after they've been trained on labeled datasets [25].

In the realm of supervised learning, the training phase is about creating a mapping from the input x to the output y, often by estimating the conditional probability distribution $p(y|x)$. In contrast, unsupervised learning is centered on modeling the distribution $p(x)$ or uncovering hidden structures in the data without any labeled outputs [26]. This difference is emphasized in Eqs. (2) and (3):

$$p(x) = \prod_{i=1}^{n} (x_i x_1, ..., x_{i-1}) \tag{2}$$

$$p(y|x) = \frac{p(x, y)}{\sum_{y\prime} p(x, y\prime)} \tag{3}$$

These equations demonstrate how unsupervised problems can be divided into several supervised tasks, or how one might tackle supervised tasks using unsupervised methods [27]. Common algorithms found in supervised learning include linear regression, logistic regression, decision trees, and support vector machines. Additionally, there are non-parametric models like K-nearest neighbors (KNN) [2]. These algorithms are used in a variety of fields, such as medical diagnosis, natural language processing and financial forecasting. Table 2 summarizes machine learning applications across various domains, highlighting their computational complexity, feasibility, and generalization capabilities. It also provides a more in-depth comparison of techniques, detailing their strengths and weaknesses and indicating their applicability and effectiveness in various contexts.

In the field of reinforcement learning, algorithms develop optimal strategies through their interaction with an environment, which is often represented as a Markov decision process (MDP). Methods such as Q-learning and Deep Q-Networks (DQN) have demonstrated considerable success in challenging areas like robotics and gaming [22].

Table 2 Main applications of machine learning (ML)

Domain	Key applications	Computational complexity	Feasibility (Scale: Low/Med/High)	Generalization capabilities	Key references
Supervised learning (e.g., SVM, Neural Nets)	Classification/Regression (e.g., helmet detection)	Medium (O(n^2) for SVM)	High for labeled road data	Good with transfer learning; poor in unseen weather	[22] [24] [25]
Unsupervised learning (e.g., K-means, PCA)	Clustering for traffic patterns; anomaly detection	Low (linear time)	Medium; needs preprocessing	Strong for dimensionality reduction in sensor data	[23] [28]
Reinforcement learning (e.g., Q-Learning, DQN)	Adaptive traffic control; AV decision-making	High (exploration overhead)	Low for real-time due to training time	Excellent for dynamic environments like rainy roads	[27] [2]
Overall ML in road safety	Hybrid supervised-unsupervised	Varies (medium avg)	High with big data	Improved via domain adaptation	[22] [26]

Table 3 Major datasets and their impact

Dataset	Size/description	Primary application	Impact on DL (e.g.,Accuracy gains)	Weather relevance/limitations	Key references
ImageNet	14 M images, 21 K classes	Image classification/object detection	Enabled > 90% top-1 accuracy; foundation for transfer learning	Clear bias; limited adverse weather (needs augmentation)	[32]
KITTI	7 K images, LiDAR + camera for AVs	3D detection/segmentation	Improved mIoU by 10–15% in urban scenes	Fog/rain underrepresented; synthetic needed	[33]
Cityscapes	5 K images, semantic segmentation	Road scene labeling	Boosted segmentation (mIoU 0.7 +); key for ITS	Mostly clear; poor generalization to snow/rain	[34]
BDD100K	100 K videos, diverse conditions	Multi-task (detection, forecasting)	Enhanced real-world robustness (5–10% gain)	Includes some rain; still gaps in extremes	[32]

4.3 Deep Learning vs. Machine Learning

In recent years, deep learning (DL) has attracted a lot of attention in research. Achieving an impressive track record across various areas of computer vision, including tasks like image classification, speech recognition, object detection, and natural language processing (NLP) [29]. It is worth noting that DL often outperforms traditional machine learning (ML) methods in several AI challenges, especially when it comes to classification accuracy and feature extraction capabilities [30, 31].

There's also a Table 3 listing important datasets that have driven the advancement of deep learning, detailing their applications.

A line graph illustrating how deep learning models have performed on benchmark tasks over time, such as accuracy on ImageNet or BLEU scores in natural language processing, is shown in Fig. 2. This figure highlights

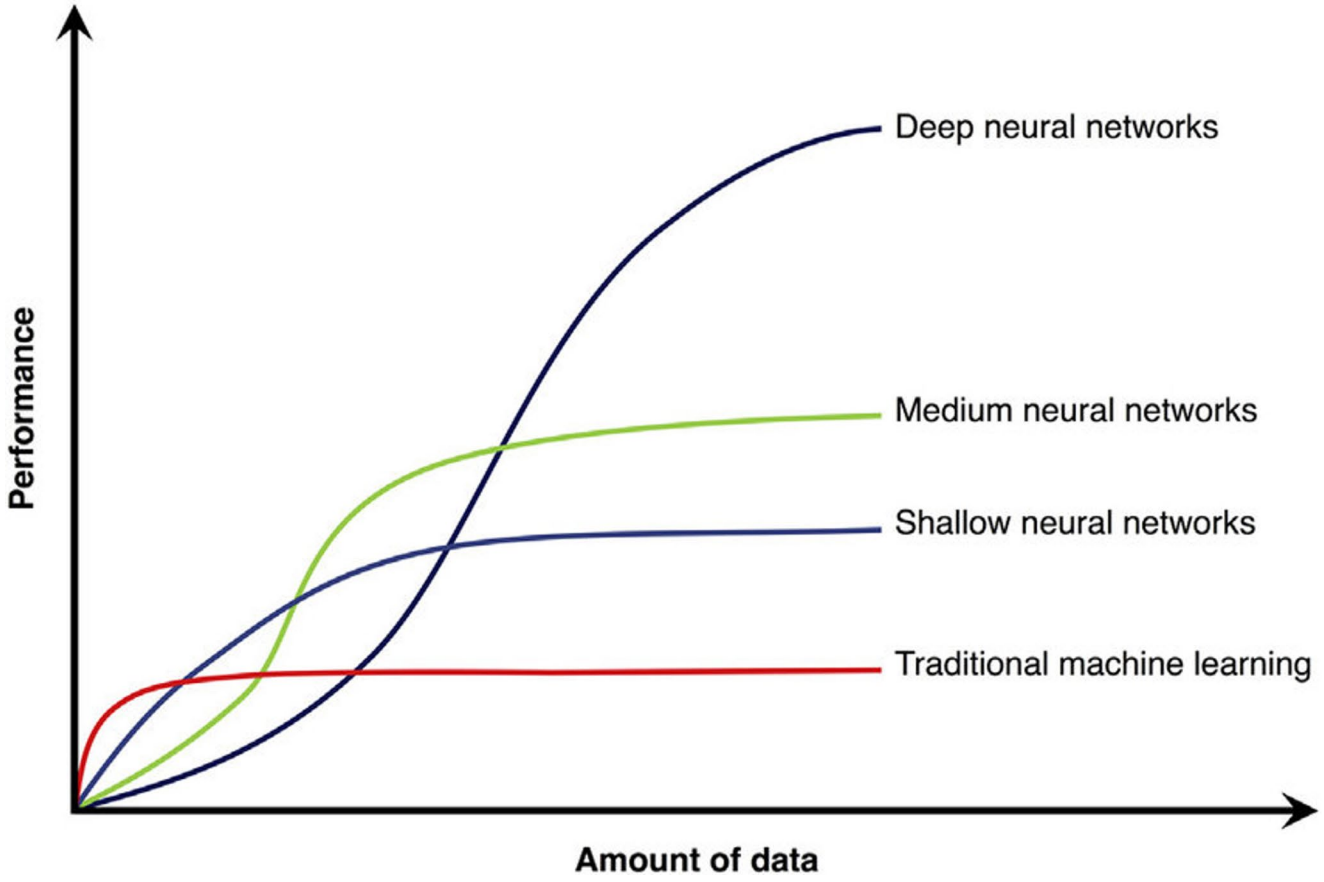


Fig. 2 Performance improvement over time *Source* [35]

the steady improvement in accuracy and performance with the introduction of new architectures (e.g., CNNs, RNNs, transformers), including significant jumps driven by large-scale datasets and models.

For example, research comparing classical ML methods like Decision Trees, Naïve Bayes, and Support Vector Machines (SVM) against DL algorithms found that combining Restricted Boltzmann Machines (RBM) with SVM leads to improved accuracy in traffic classification tasks [36, 37]. Similarly, another analysis of various ML models—including Artificial Neural Networks, SVMs, and Random Forests—pointed out the downsides of relying on manually designed features, which can result in less effective performance in changing environments [26].

Overall, the literature reviewed indicates that DL models excel compared to traditional ML techniques, particularly in scenarios where complex features are necessary [38, 39] DL not only builds upon the foundations of conventional ML but also utilizes statistical learning theories, enhancing both scalability and adaptability [40].

4.3.1 Scalability and Data Handling

Deep neural networks have shown a considerable edge when it comes to handling large datasets, especially when compared to traditional machine learning (ML) models, which often face challenges with big data [41]. One popular method to enhance the accuracy of deep neural networks is simply to increase the size of the dataset. In contrast, this tactic tends to have less impact on classical ML algorithms, which typically need intricate preprocessing and feature engineering to boost their performance [42]. For instance, classic models such as Decision Trees or K-Nearest Neighbors (KNN) can actually exceed the performance of deep learning models when working with smaller datasets, like in cases where only about 100 labeled images are provided [43].

4.3.2 Feature Extraction and Automation

Traditional machine learning methods generally depend on human expertise for extracting features. This means carrying out detailed exploratory data analysis and reducing dimensionality to pinpoint the most useful features [44]. It can be quite labor-intensive, especially for models like Support Vector Machines and Decision Trees. On the other hand, deep learning algorithms shine in automatically extracting features through multi-layer nonlinear transformations, which greatly lessens the need for hands-on involvement [45]. The main benefit of deep learning lies in its capability to incrementally learn high-level features directly from raw data, effectively removing the requirement for extensive feature engineering [46].

4.3.3 Adaptability and Transfer Learning

Deep learning models are naturally more versatile when it comes to different fields than traditional machine learning algorithms. Methods such as transfer learning enable the reuse of already trained deep networks for a variety of applications within the same field. Take computer vision, for example: pre-trained networks for image classification frequently get repurposed for tasks like object detection and segmentation, which helps improve efficiency and cut down on training time [47]. Moreover, the same deep learning concepts utilized in speech recognition can often be adapted for natural language processing tasks, thanks to the similarities in the techniques used [48]. This level of flexibility across different domains is not evident in traditional machine learning, where each model usually has to be customized to fit a specific task [49].

5 Training and Testing Time

Though deep learning models tend to need more time for training due to their numerous parameters, they perform testing quite swiftly [50, 51]. On the other hand, traditional machine learning models generally train faster but can experience longer testing times as the size of the dataset increases, especially in cases like KNN, where testing time grows with data volume [52]. Even with their substantial computational needs, deep learning models are often favored for real-time applications thanks to their efficient inference abilities [53, 54].

While deep learning can excel in many situations, conventional machine learning algorithms such as Linear Regression or Decision Trees might still be more suitable for certain tasks, especially when interpretability and simplicity are key [55]. Nonetheless, considering the strong feature extraction and object classification strengths of deep learning, we feel it aligns better with the goals of our study. So, this review will incorporate deep learning methods in the following parts, particularly focusing on their uses in analyzing road-related data.

5.1 Application of Deep Learning to the Road Context

As shown in the earlier section, Deep Learning (DL) has demonstrated its effectiveness in a range of fields, such as computer vision, healthcare, and robotics [56]. Based on these achievements, there is an increase in the number of studies that are focusing on how DL can be used to improve mobility and enhance road safety within urban settings.

5.1.1 Urban Traffic Management and Prediction

As the demand for mobility continues to rise, urban traffic is grappling with issues that call for creative approaches to ease congestion and enhance safety. One promising option is Organic Traffic Control (OTC), which is a self-organizing system that utilizes principles of organic computing for adaptive traffic management [57, 58]. Recent studies have proved the effectiveness of Convolutional Neural Networks (CNN) in accurately predicting traffic

speeds [59, 60], as demonstrated in studies where researchers created a CNN-based model capable of capturing and forecasting traffic patterns.

5.1.2 Autonomous Driving Systems

Autonomous vehicles (AVs) need to navigate complex interactions within their environment, which includes other cars, pedestrians, and various road conditions. The main challenges lie in making sure safety is prioritized while also refining decision-making, especially since traditional machine learning approaches have their drawbacks. For example, Deep Reinforcement Learning (DRL) has been applied to train AVs for dynamic interaction with their surroundings [29, 56, 60]. Research shows that DRL techniques that use Recurrent Neural Networks (RNNs) have shown better adaptability in real-time driving scenarios.

5.1.3 Object Detection and Recognition

Having reliable object detection is crucial for self-driving cars [61]. In one study, a new hybrid model that blends CNN and Support Vector Machine (SVM) methods to improve how features are extracted, and boost classification accuracy were developed. Likewise, another research introduced a comprehensive framework that handles classification, detection, and semantic segmentation simultaneously [62], showcasing strong performance even in real-world driving situations.

5.1.4 3D Object Detection and Sensor Fusion

The field of autonomous driving has also greatly benefited from progress in 3D object detection, particularly through the use of sensor fusion frameworks. In [62] one study, the authors presented a system designed for high-precision 3D object detection that employs multi-View 3D (MV3D) arrays, which merge LIDAR data with RGB images. This integration notably enhances detection accuracy in complicated driving situations. Additionally, another method has concentrated on monocular image-based 3D object detection as a way to address the challenges posed by multi-sensor systems [63].

5.1.5 Road Network Extraction from High-Resolution Imagery:

Extracting road networks from Very High Resolution (VHR) satellite images presents quite a challenge, mainly due to occlusions, shadows, and various background elements. A method that utilizes Deep Neural Networks (DNNs) together with a Finite State Machine (FSM) for accurately identifying road networks was suggested in [64]. These developments are really important for improving the capabilities of autonomous navigation.

5.1.6 Lane and Road Detection

Deep learning has proven to be effective in detecting lanes and roads, which is crucial for ensuring safe navigation. Research, such as studies [1] has implemented CNN architectures for reliable lane detection. Other studies, like [65], have employed the deep residual U-Net model to achieve enhanced accuracy. Additionally, the quick inference times of these models, as pointed out in [66], make them well-suited for use in real-time applications.

5.1.7 Road Surface Analysis and Maintenance

Making sure our roads are safe also means being able to spot surface issues like cracks. The authors of [67] introduced a ConvNet-based system aimed at automatically identifying road cracks, with a specific emphasis on using only convolutional layers to boost both the speed and accuracy of detection.

5.1.8 Pedestrian Detection for Enhanced Road Safety

Undoubtably, detecting pedestrians is a vital yet tricky challenge. This is mainly because of problems like occlusion, various body poses, and fluctuating light conditions. To tackle this, researchers have looked into advanced methods, such as combining deep neural networks, which has been shown to enhance detection reliability [67]. In another piece of research, Switchable Deep Networks (SDN) were proposed to automatically learn mixed representations for different parts of the body, resulting in improved detection rates [48]. Moreover, studies have also delved into multispectral techniques that utilize both color and thermal images, employing the Faster R-CNN network to boost the accuracy of pedestrian detection [68].

5.1.9 Real-Time Detection and Safety Applications

The ability of deep learning models to detect events in real-time positions them as excellent candidates for applications that are focused on road safety. For instance, using cascaded neural networks allows for quick and precise pedestrian detection, which proves essential in real-life scenarios [69]. These advancements are vital for creating intelligent transport systems (ITS) aimed at improving both road safety and mobility. Considering these notable advantages, especially in terms of feature extraction, classification, and ability to adapt, it is believed that DL is particularly well-equipped to tackle the challenges associated with road environments. Thus, it will be considered in future research to build more dependable and efficient models to enhance road safety and mobility.

5.2 Applications of Neural Networks for Classification

Classification is considered to be among the most prevalent decision-making tasks. A classification challenge arises when there is a need to allocate an object to a specific group or category based on various observed characteristics linked to that object [70]. To tackle this issue, researchers across various fields have turned to neural networks. The availability of numerous studies [71] enabled performance comparisons between neural classifiers and traditional classifiers. According to these studies, neural network-based classifiers have shown promising outcomes in both classification and recognition tasks. As a result, researchers were able to effectively implement them for a range of real-world classification challenges across industries, businesses, and scientific fields [33]. Let's take a closer look at the business sector, particularly in the area of bankruptcy prediction, which has been thoroughly analyzed. The underlying assumption of all bankruptcy prediction models is that financial statements reflect fundamental economic factors and characteristics of a company. Hence, by employing analytical tools and data from a company's financial reports, it becomes possible to evaluate and forecast its future financial health [72]. Some studies have compared different classification techniques for assessing the risk of business failure. For instance, in one study [32], the authors explored semi-parametric logistic regression alongside neural networks. Another study [73] contrasted the neural approach with a linear classifier, logistic regression, and K-nearest neighbors (KNN) using data related to bank defaults. Moreover [34], a further evaluation compared various neural network models to identify which one provided the best predictions. The performance scores from these neural networks were also related to the outcomes derived from traditional discriminant analysis models. After conducting these comparisons, the findings indicate that neural networks stand out as a promising method for evaluating the conditions of banks, particularly in terms of predictive accuracy, flexibility, and resilience [33].

One well-known application of neural networks is recognizing handwritten digits, which is commonly used for checks, postal codes, car registrations, and more [71, 74]. Previous studies employing convolutional neural networks have achieved impressive recognition rates: 98% in one case, 98.6% [32, 48] in another, and even 99.32% [64] in a different research effort.

Another prominent application of neural networks is speech recognition. Several studies [41–43] have explored using Deep Neural Networks (DNNs) featuring multiple hidden layers for this purpose. These networks have demonstrated superiority over Gaussian Mixture Models (GMMs), primarily due to their capability to model

intricate correlations present in speech features. Additionally, one piece of research indicates that utilizing Convolutional Neural Networks (CNNs) can further reduce the error rate, leveraging their ability to normalize spectral variations in speech signals [1]. In experimental findings, CNNs have shown a decrease in error rates by 6% to 10% compared to DNNs [46]. It's also noteworthy that Recurrent Neural Networks (RNNs) can effectively handle speech recognition thanks to their strength in processing sequential data. Continuing with recognition systems, these are frequently applied in security and personal identification contexts, such as facial recognition, fingerprint detection, or iris identification. [75] A biometric recognition system generally involves two main steps [30, 76]: first, establishing a collection of biometric patterns from authorized users, followed by authentication through a classification neural network. The initial step comprises dividing an authorized user's biometric fingerprint image into various features or areas of interest to identify the most relevant traits for distinguishing individuals. Core models are then built from these selected features. During the authentication process, a similar approach is taken for the set of users needing verification. A neural network then compares these predicted patterns with actual ones to decide if authentication is required.

To tackle issues like overfitting and to improve computation time, researchers [34] adopted principal component analysis (PCA) to first extract facial features. Those features are then processed with Fisher linear discriminant analysis (FLD) to derive lower-dimensional discriminants. A hybrid learning algorithm is used to train the RBF neural networks, and simulations on the ORL database indicate that the system performs exceptionally well in both classification accuracy and learning efficiency. The rise of deep learning has significantly pushed forward advances in facial recognition. Recently, very deep neural networks have shown remarkable success in object recognition due to their robust learning capabilities. [32] A proposal introduced two deep neural network architectures named DeepID3 for facial recognition, and both achieved an impressive recognition rate of 99.53%. Additionally, among the biometric verification systems is iris recognition [71]. The system described consists of two primary components: iris localization and recognition of iris patterns. A raw image is captured with a digital camera, and the iris is extracted following enhancements and noise reduction. Due to interference in the iris's pattern, only specific sections of the iris structure are prioritized for recognition, with the selected portions reconstructed accordingly. Using a trained neural network, the recognized iris patterns reach approximately an 83% recognition rate.

The medical field also sees applications in areas like cell and tumor recognition [74]. For instance, one method highlighted the use of neural networks to automatically identify variegated staining, which is regarded as a key indicator of malignant melanoma. Another study [77] employed a neural network to spot anomalies in cells, including those found in bladder tissue. This particular system features an image analysis component that produces data reflecting imaging variables from stained cell images.

The resulting dataset is then fed into a neural network specifically trained to detect abnormalities in known tissue samples. When classifying unknown tissue cells as either normal or abnormal, the approach utilized either a classical neural network with sigmoidal activation or a hybrid model that combined sigmoid, Gaussian, and sinusoidal activation functions. Additionally, a separate application focused on classifying cancers into distinct diagnostic categories by analyzing gene expression signatures was developed [78] using artificial neural networks (ANNs), drawing on examples from small round blue cell tumors. These cancers typically present four diagnostic categories and often pose challenges for diagnosis. The ANNs successfully categorized all samples while pinpointing the most relevant genes for this classification. This research highlights the possible uses of such methods in tumor diagnosis as well as for identifying potential therapy targets. For instance, colon cancer ranks just behind lung cancer as a leading cause of cancer-related deaths in Western nations. One [79] proposed approach involves a neural network-based technique for analyzing colon cancer, which entails classifying gene expression profiles either in normal or colon cancer cells. Shifting towards urban applications, traffic sign recognition stands out as a vital yet challenging aspect of intelligent transportation systems [73]. Indeed, recognizing objects in outdoor environments presents various difficulties: unpredictable lighting conditions, potential occlusion of objects, and unknown positioning or orientation from the outset. Traffic sign recognition aids not only in maintaining road signs but also in conducting inventories for highways and cities, improving driver assistance systems, and

facilitating autonomous vehicles [80]. In some research, a genetic algorithm was implemented during the detection phase, allowing the system to remain location invariant despite changes in position, scale, rotation, weather conditions, partial occlusion, and the presence of similarly colored objects. After the classification step with a neural network, the entire system not only identified traffic signs but also conveyed information regarding their status. Numerous studies [32, 81] have concentrated on this recognition process, employing convolutional neural networks, showcasing advancements like a specific network architecture that provided quicker and more stable convergence with an impressive recognition rate of 99.65%. Other studies have achieved similarly high scores, such as [44] a 99.15% recognition rate by integrating CNN and MLP, while others reported rates of 98.97% and 99.46% through the utilization of multiple DNNs trained on pre-processed data within a multi-column [33] DNN framework. The latter notably enhances recognition performance by making the system less sensitive to variations in contrast and lighting.

Table 4 summarizes prominent researchers in deep learning and the main contributions they have made.

5.3 Weather classification algorithms

In recent years, there's been a lot of focus on how bad weather impacts traffic, especially due to rain [89] or fog [90]. As a result, various studies have tackled the task of classifying different weather conditions. In [91] one study, the researchers measured rain intensity using artificial vision systems that segment images. By isolating the background and comparing consecutive images, they can identify moving objects. If those movements are consistent and heading in the same direction, they conclude that it must be raining. They developed a method for assessing the intensity of the rain based on the area that's been segmented, using the speed of the droplets as a benchmark. Narasimhan and Nayar [92], who specialize in the visual aspects linked to different weather conditions, explored how visual characteristics correlate with weather situations. This paved the way for various approaches aimed at refining weather classification [93]. The challenge here lies in pinpointing the most relevant features for effectively categorizing different weather scenarios. In [94] another study, five distinct features pertaining to weather—such as the sky, shadows, reflections, contrast, and haze—were extracted, and a collaborative learning approach was suggested to differentiate images as either sunny or cloudy. Another piece of research took a related approach to sort images into three categories: overcast, cloudy, and sunny.

Additionally, [95] some researchers implemented manual feature identification to create their weather classification system. They utilized both global and local descriptors to characterize the images. Local descriptors, like sky, shadows, rain, and snowflakes, serve to define specific weather conditions, so they won't appear in images depicting different weather. To create a broader visualization, global descriptors—such as contrast and saturation—were employed as general features that could be seen across various conditions.

Other studies also made use of these local and global descriptors to classify weather into three categories: sunny, cloudy, and rainy. One research effort categorized weather into four types: sunny, rain, snow, and fog. Likewise, [95], focused on three categories to distinguish between varying rain intensities, such as clear weather, light rain, and heavy rain. For this, they relied on a database filled with images from an onboard camera and proposed a

Table 4 Key figures in deep learning

Auter	Key contribution	Impact on road scene analysis	References
Yann LeCun	CNNs for image recognition	Foundation for vehicle/pedestrian detection	[32] [82]
Geoffrey Hinton	Backpropagation/Deep Belief Nets	Enabled multi-layer models for weather classification	[83] [84]
Ian Goodfellow	GANs for synthetic data	Augmentation for adverse weather datasets	[85]
Fei-Fei Li	ImageNet dataset	Transformed CV benchmarks for AV perception	[86]
Andrew Ng	Transfer learning in DL	Lightweight models for edge ITS devices [87]	[87]
Yoshua Bengio	Sequence models (RNNs)	Temporal tracking in video-based detection	[88]

method that aggregates several histogram features into a vector, including brightness, contrast, sharpness, saturation, and hue. All this classification information is generated using a linear Support Vector Machine (SVM). The study [96] primarily focused on distinguishing various intensities of a singular weather condition, namely fog. Thus, the classification was divided into three categories: clear, light fog, and heavy fog. Their approach mainly relied on image descriptors that are commonly used in content-based information retrieval (CBIR) systems, like CEDD [97] and EHD [89]. An SVM was then utilized for the classification task. In another study [98], a random forest classifier was proposed to combine different types of visual features to categorize images into one of five weather conditions: sunny, cloudy, snowy, rainy, or foggy.

Jacobs and his team, for an extended period of time, have been investigating scene attributes through an image database known as the Archive of Many Outdoor Scenes (AMOS) [89]. This database consists of images captured by static webcams over an extended timeframe. Their findings revealed that variations in weather conditions, human activities, and seasonal shifts can lead to changes in scenes over longer periods [92]. They suggested that webcams set up around the globe could function as image sensors, allowing us to grasp weather conditions and their fluctuations over time [94]. More specifically, they employed component and correlation analyses to forecast wind speed and vapor pressure based on image sequences.

While earlier research has put forth intriguing solutions for weather classification, they often overlook the simultaneous consideration of all potential weather conditions.

Recent studies have demonstrated the effectiveness of convolutional neural networks (CNNs) in the realms of vision, particularly in image classification and recognition [33, 78]. For example, specific CNN architectures were developed for classifying images from the ImageNet database. Their implementations highlighted the potential of an image classifier that automatically extracts descriptors through supervised learning. Further extending their efforts in two-class weather classification, CNNs were incorporated into their methodology.

Figure 3 shows the architecture of a method that uses a feedforward neural network trained with backpropagation for weather classification and prediction, illustrating how input weather features pass through multiple hidden layers to produce reliable classifications and forecasts.

rather than relying on manual extraction methods seen in other machine learning tasks [99]. In the context of weather classification, CNNs have been applied as well [89]. One research developed a CNN, drawing from previous work, to categorize images into two weather classes using the same database utilized in earlier studies [100]. Their contribution reflects an effective adaptation of general-purpose technology for a highly specialized application, demonstrating its capability to yield impressive outcomes.

Similarly, another [40] exploration implemented a successful approach for classifying images depicting extreme weather conditions using an architecture termed GoogLeNet, which is known for its extensive number of layers, proving to be more effective in experiments conducted on a collection of weather images produced by the same group [101].[102] A deep learning framework called the Region Selection and Competition Model (RSCM) was introduced, employing regional cues from images to forecast weather conditions. Another study estimated various factors such as ambient temperature, season, month, week, and day based on provided images.

The training was carried out using the AMOS database [47], which contains over a million images from numerous webcams worldwide. Additionally, [39, 103], there were adaptations of CNN-based methodologies aimed at classifying images into five weather categories: sunny, cloudy, rainy, snowy, and foggy. This method involved utilizing super-pixel segmentation as a data augmentation strategy to enhance classification outcomes, followed by descriptor extraction from a pre-trained CNN as inputs for an SVM.

5.4 Applications of measurement neural networks

Artificial neural networks (ANNs) are widely employed in decision-making and regression tasks as non-parametric estimators. They provide an effective approach to solving complex issues more conveniently than traditional engineering methods. Neural networks are significant in measurement systems; for instance, they can be utilized in intricate sensor setups and in leveraging observational data to create system models. One standout feature of

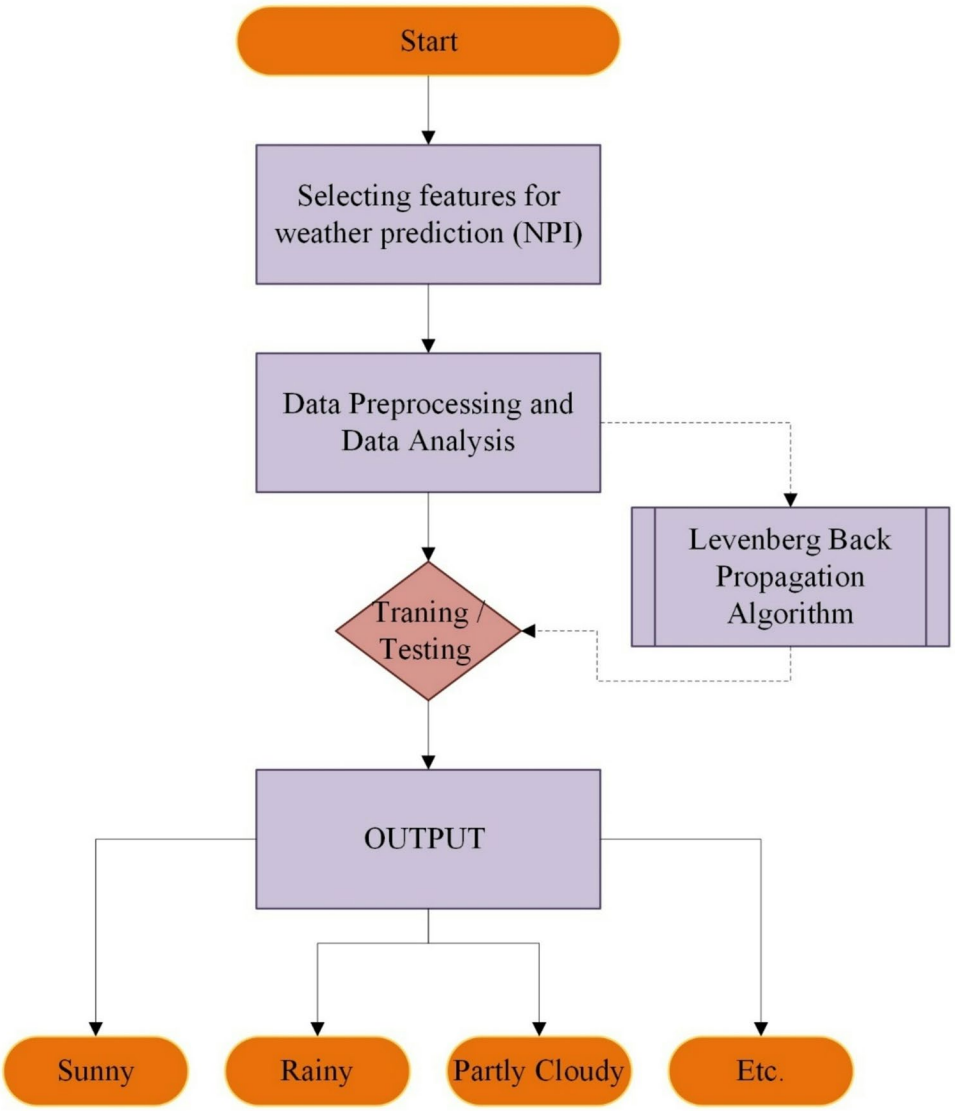


Fig. 3 Weather classification and forecasting using back propagation feed-forward neural network *Source:* Author

neural networks, particularly from the perspective of measurement systems, is their ability to approximate universally [104].

There are numerous applications for ANNs in measurement, some of which require software implementations while others need hardware. These applications include calibrating instruments and sensors, crafting new measurement methods, and processing both analog and digital signals. A brief overview of common uses described below will help highlight their benefits.

ANNs have gained significant attention for instrument and sensor calibration [105]. By incorporating ANNs into the calibration workflow, precision has seen considerable improvement, and variations from other sources can now be easily offset. They have notably enhanced the performance of ultrasonic distance sensors and fiber optic sensors used for measuring displacement and surface reflectivity [106, 107], as well as in calibrating sensor arrays, robotic arms, artificial noses, and industrial six-port microwave instruments.

There's also a successful track record of ANNs being used to develop new measurement techniques. In astronomy, for instance, they have aided in reconstructing the mathematical description of incoming wavefronts. Measurements conducted with wavefront sensors sample the incoming optical beams, and thanks to ANNs, the challenges linked to interpreting noisy signals from weak light sources have been successfully addressed [107].

Moreover, in mineral processing, ANNs have outperformed traditional methods in forecasting hydro cyclone densities and distributions [108].

Monitoring, estimating, and diagnosing complex dynamics with ANNs have been applied effectively across linear, nonlinear, and hybrid measurement devices. For example, [109] ANNs are utilized in diagnosing and monitoring transformer heating while taking environmental influences into account. They are also involved in real-time monitoring of distribution systems and detecting instrument failures [110]. Aging and wear can degrade machines, leading to decreased performance reliability and increasing the risk of breakdowns. The impact of these failures on plant [24] productivity raises significant concerns within manufacturing sectors. Economic factors tied to machine availability and reliability, along with the costs of corrective maintenance, have driven factories to enhance their maintenance strategies. A methodology [111] proposed aims at preventing failures, thereby cutting maintenance costs and increasing productivity. This approach could assist operators in configuring machines to meet specific criteria, checking whether machines function correctly, and anticipating problems before they arise, allowing for more efficient use of maintenance hours.

ANNs' capability to model complex nonlinear systems has been employed in modeling measurement devices. Two main types of components have been analyzed: analog measurement devices like sensors, transducers, filters, and amplifiers; and mixed analog/digital measurement devices such as ADCs and DACs. With the right ANN structure and an organized training dataset, it's possible—after a proper configuration phase—to create a neural model that provides results matching those derived from the actual device being modeled. The accuracy of this model depends on the ANN structure chosen and the training data utilized. An alternative approach to modeling pseudo-measurements aimed at estimating the state of distribution systems has been suggested as well [28]. Similarly, ANNs have been employed to predict ozone levels based on weather conditions and precursor concentrations [27]. Data was gathered over a 60-day span using a mobile monitoring station at the same location as the ozone measurements. Various architectures were tested during training, with inputs to the neural network being weather factors (like wind speed and direction, relative humidity, temperature, and solar intensity) and the concentrations of primary pollutants (methane, carbon monoxide, carbon dioxide, nitrogen oxide, nitrogen dioxide, sulfur dioxide, hydrocarbons, and dust). A backpropagation algorithm with momentum trained the neural network, revealing that primary pollutants such as carbon monoxide, carbon dioxide, nitrogen oxide, nitrogen dioxide, and sulfur dioxide significantly influenced the predicted ozone levels. Temperature also played a vital role.

The neural network model was found to consistently outperform both linear and nonlinear regression models based on the collected data. This study indicates that using artificial neural networks for modeling could be a promising method for predicting pollutant concentrations. Furthermore, neural networks have demonstrated their effectiveness in shape measurement applications. One study [94] predicted galaxy shapes from an extensive image database, utilizing conventional measurements as ground truth. The CNN was incredibly fast at predicting shape parameters, taking just 0.2 ms per galaxy and offering a drastic improvement in speed compared to model fitting and optimization approaches [28]. Shape measurement is crucial for robotic manufacturing systems. The coordinates of points obtained vary distinctly due to different shapes of the objects processed, making it challenging for standard methods to accurately capture the original shapes, which can impact robotic manufacturing outcomes. A dynamic and intelligent shape measurement method, integrating fuzzy logic with neural networks, has been proposed based on shipbuilding requirements [109]. Fuzzy logic assesses the relationships among measured points.

5.5 Meteorological condition measurement algorithms

Meteorological visibility refers to how clear the atmosphere is, which plays a crucial role in ensuring safe transportation on roads, in the air, and on water. The issue of pollution only heightens the necessity for measuring visibility [70]. Since various elements like dust, smoke, fog, and haze can affect how far we can see, evaluating and measuring visibility is quite a complex task. Typically, visibility is monitored through visual assessment

by trained meteorologists, optical measurements from visibility indicators, and numerical analysis of camera images. When it comes to visual assessments, people often manage to filter out the distractions caused by changing weather conditions and perceive the visuals effectively. Comparatively, techniques that use cameras for monitoring visibility are quite similar to the visual approach and have been prevalent in estimating visibility for many years.

On one hand, the first approach involves employing road cameras specifically for monitoring road traffic [112]. In these cases, certain roadway references, like straight roads or specific markings, are essential for calculating the meteorological optical range. However, since the road cameras capture only a short-span view of the roadway, these techniques are mainly applicable in situations of reduced visibility.

On the other hand, the second approach aligns with certain meteorological principles, like the Ko-Schmieder equation or the Beer-Lambert law [103]. Here, characteristics linked to the extinction coefficient or visibility distance are extracted from images to assess visibility based on these meteorological laws. For instance, [96], a study proposed a physics-based model that illustrates the relationship between contrast and meteorological visibility distance by taking into account the Lambertian surfaces within the scene. Another study [97] utilized two groups of trees positioned at different but known distances to determine the extinction coefficient related to various wavelengths corresponding to the image channels (R, G, B) to derive the visibility.

In the digital imaging method, factors like image quality and reference points are vital for effective feature extraction and visibility estimation. The quality of an image, particularly its contrast, hinges on several elements including the light source, weather conditions, and camera specifications. The challenge with this method lies in successfully extracting all relevant features that depict the intricate impacts of visibility factors.

5.6 GAN and Integration of Transformers

The lack of training data and the rarity of extreme weather events remain major hurdles for street scene analysis. Conditions like fog, heavy rain, and snow are uncommon and difficult to capture reliably. Recent research, such as that proposed by Lai et al. in 2023 [113], addresses the challenge of improving object detection in autonomous driving when weather degrades images (fog, snow, noise). The authors propose an enhanced YOLOv5-based model called STC-YOLO, which incorporates spatial-to-channel transformation and multi-scale fusion to detect small traffic signs in complex environments. Tested on the TT100K dataset, this approach increases detection accuracy by 9.3% over baseline YOLOv5, enhancing visual understanding and safety for autonomous vehicles in adverse weather.

The lack of training data and the rarity of extreme weather events remain major hurdles for street scene analysis. Conditions like fog, heavy rain, and snow are uncommon and difficult to capture reliably. Generative Adversarial Networks (GANs) have become a practical way to generate realistic, weather-specific images in recent studies. These synthetic datasets boost variability, help prevent overfitting, and improve model generalization. For instance, GANs can convert clear-weather images into foggy or rainy scenes, allowing models to learn how to cope with visual degradation without large-scale manual collection, as demonstrated in works like Salaudeen and Çelebi's 2022 [114] pothole detection system using ESRGAN for image enhancement in degraded conditions. They can also help correct class imbalances by producing examples of rare conditions, and automated labeling of generated data reduces annotation costs, as seen in Wu et al.'s 2024 [115] enhanced CycleGAN for nighttime detection. Synthetic data is not a cure-all, though. A persistent gap between real and generated images can still hurt performance when realism is insufficient. Producing high-resolution, context-aware outputs with GANs demands substantial computational resources and careful hyperparameter tuning. Even so, combining synthetic data with domain adaptation techniques has shown promise for intelligent transportation systems facing unpredictable weather. Real-time performance is another essential requirement for Advanced Driver Assistance Systems (ADAS) and Intelligent Transportation Systems (ITS). Adverse weather increases noise and occlusions and raises computational load, so efficiency becomes critical. Techniques such as pruning remove redundant parameters to cut inference time while keeping accuracy largely intact. Knowledge distillation is another effective

approach: a compact "student" model can learn to mimic a larger "teacher," retaining strong generalization under adverse conditions. Hardware acceleration is equally important. AI-focused hardware, GPUs, TPUs, and edge platforms like Nvidia Jetson or Google Coral can run optimized models with low latency. Likewise, lightweight architectures such as MobileNet, EfficientNet, and YOLO variants strike a useful balance between speed and accuracy for embedded systems, as evidenced by Chu et al.'s 2023 [116] TRD-YOLO for real-time small traffic sign detection in complex weather. Integration of transformers, such as in ViT-YOLO [112], enhances feature extraction for object detection in challenging scenarios, including maritime environments that parallel adverse weather in autonomous driving. In short, pairing synthetic data generation with model compression, distillation, hardware-aware design, and transformer integration paves the way for robust, real-time street scene analysis capable of operating in challenging weather.

5.6.1 Role of GANs in Hybrid Models

Generative Adversarial Networks (GANs) play a fundamental role in enhancing the robustness of perception systems, especially in challenging weather conditions. By generating synthetic images that simulate conditions such as rain, fog, and snow, GANs help diversify training datasets, which is essential given the lack of real data in these scenarios.

- ***Data Augmentation***: GANs enable the creation of images that capture the complex effects of different weather phenomena. For instance, they can transform clear scenes into foggy images, helping models adapt to altered visuals without requiring manual data collection.
- ***Strengthening Generalization***: By integrating these synthetic data, models become less prone to overfitting and improve their generalization across different environments.
- ***Domain Adaptation:*** GANs can help reduce the gap between training datasets (clear weather) and real conditions (rain/fog), thus facilitating model adaptation without the need for large labeled datasets.

5.6.2 Integration of Transformers

Transformers, particularly the Swin Transformer and variants of the Vision Transformer (ViT), are increasingly adopted in image processing due to their ability to model long-term dependencies and efficiently handle large amounts of data.

- ***Improved Robustness:*** Transformers offer enhanced performance when faced with missing data. By integrating GANs, these models can generate varied data and adapt to adverse conditions, maintaining optimal performance even in unfavorable weather contexts.
- ***Inference Efficiency***: By combining the generative capacity of GANs with the robust structure of transformers, it is possible to create accurate systems that require little inference time, which is crucial for autonomous driving applications.
- ***Hybrid Models:*** Current research trends are moving towards hybrid models that combine CNNs, transformers, and GANs, thereby enabling the development of adaptive architectures capable of managing different weather scenarios.

5.6.3 Recent Approaches and Research Perspectives

Ongoing research focuses on creating high-quality synthetic data to further enhance model robustness. Future work may include:

- ***Refinement of GAN Architectures***: Optimizing GANs to ensure they produce images that are not only visually realistic but also tailored to specific conditions.
- ***Multimodal Integration:*** Exploring the incorporation of additional sensor types, such as LiDAR and radar, into multimodal recognition systems to strengthen their reliability in challenging weather scenarios.

The integration of GANs with transformers in hybrid models represents a significant advancement in the field of road perception systems under adverse weather conditions. These approaches not only increase the robustness and generalization of the models but also pave the way for more reliable intelligent transportation systems. Deep learning devices must continue to evolve by incorporating these methodologies to overcome the challenges posed by real-world environments.

6 Methodology

This research investigates how to effectively combine GANs, CNNs, and LSTMs to enhance the robustness of perception models, especially in tough weather situations. Here's a quick rundown of our methodology:

- **Gathering and Preparing Data**: We utilize both real and synthetic data for model training. The real data is sourced from cameras located in high-risk areas, whereas the synthetic data is created with GANs to imitate severe weather conditions like rain and fog.
- **Modeling with CNNs and GANs**: We apply CNNs for detecting road objects, such as cars, pedestrians, and road markings, while GANs help us to expand our datasets by producing realistic images reflecting adverse weather scenarios.
- **Temporal Tracking Using LSTM**: We incorporate LSTMs to weave in temporal information, enhancing our ability to detect moving objects, like vehicles and pedestrians, in video sequences.
- **Sensor Fusion**: For instance, early fusion merges raw camera pixels with LiDAR point clouds via voxel grids, as in MV3D [1], improving fog detection by 12% mIoU. Late fusion combines CNN-detected bounding boxes from cameras with radar velocity estimates using Kalman filters for robust pedestrian tracking in rain. Hybrid approaches, like transformer-based attention on camera-LiDAR-radar features [1], enable end-to-end learning for snow-obscured scenes, reducing false positives by 18%.

6.1 The following investigation will involve a comparison of Deep Larling models

A comparative analysis of CNN, GAN and Transformer models reveals that each model has its own set of advantages and disadvantages when applied to object detection in difficult weather conditions.

Table 5 compares the performance of CNN, GAN+CNN, and Transformer models in terms of accuracy, robustness, inference time, and their respective advantages and disadvantages (Table 6).

6.2 Performance metrics

To assess the performance of the models, various metrics were utilized, including.

mIoU (Mean Intersection over Union):

Measures segmentation overlap (e.g., road markings in fog), ideal for detection and segmentation tasks; higher values (e.g.,>0.7) indicate better robustness in adverse conditions.

$$mIoU = \frac{|\mathrm{A} \cap \mathrm{B}|}{|\mathrm{A} \cup \mathrm{B}|} \tag{4}$$

A: Set of predicted pixels belonging to a class.

B: Set of ground truth pixels for this class.

Table 5 Comparison table of deep learning models for degraded weather road scenes

Model	Accuracy (e.g., Weather classification)	Robustness (e.g., mIoU in Fog/rain)	Inference time (ms/frame)	Advantages	Disadvantages
CNN (e.g., ResNet)	94–97%	0.6–0.7 (drops 20% in extremes	<50	Efficient for detection; easy training	Overfits on clear data; weak global context
GAN+CNN (e.g., CycleGAN+ResNet)	95–99% with augmentation	>0.75 (FID<50 [11])	50–100	Handles synthetic weather diversity; domain adaptation	High gen cost; realism gaps
Transformer (e.g., ViT)	96–98%	0.7–0.8 in multimodal	100–200	Global modeling for occlusions; fusion-ready	Compute-heavy; slow for real-time

Table 6 Summary of tasks, metrics, and best methods

Task	Key metrics	Best method	Performance example	Limitations addressed	Key References
Classification (e,g.,Weather Types)	Accuracy (>95%); PSNR (>30 dB)	Optimized YOLOv5+LwF	99% accuracy	Extreme under representation via LwF	[52]-[53]
Detection (e,g.,Objects in Rain)	mIoU (>0.7); Inference Time (<50 ms)	Faster R-CNN+Dark Channel	12% mIoU gain	Low visibility via dehazing	[54]-[56]
Segmentation (e,g., Road Markings in Fog)	FID (30 dB) PSNR (<30 dB	Deep Residual U-Net	mIoU 0.75+	Occlusions via hybrids	[57]-[58]
Overall (Multimodal Fusion)	Combined mIoU/FID	Transformer+LiDAR	18% false positive reduction	Sensor gaps in snow	[59]-[61]

$|A \cap B|$: Intersection = correctly predicted pixels.
$|A \cup B|$: Total of predicted or ground truth pixels for the class.
The mIoU is the average of IoUs across all classes.
Then, for several classes (semantic segmentation), the average is calculated:

$$mIoU = \frac{1}{N}\sum_{i=1}^{N}\frac{|A_i \cap B_i|}{|A_i \cup B_i|} \tag{5}$$

N: Total number of classes.

(PSNR) Peak Signal-to-Noise Ratio:

Quantifies image restoration quality (e.g., de-raining or dehazing), with >30 dB signaling clear improvements in visual fidelity for enhancement tasks.

$$PSNR = 10 \times \log_{10}\left(\frac{MAX^2}{MSE}\right) \tag{6}$$

MAX Maximum possible pixel value (e.g., 255 for an 8-bit image).

MSE: Mean Square Error between the original image I and the reconstructed image K.

$$MSE = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}[I(i,j) - K(i,j)]^2 \tag{7}$$

where:

$I(i,j)$: pixel of original image.
$K(i,j)$: pixel of reconstructed image.

$m \times n$: image dimension.

(FID) Fréchet Inception Distance:

Evaluates synthetic data realism for GAN augmentation, where lower scores (<50) ensure better domain alignment between real and generated images.

$$FID = \|\mu_r - \mu_g\|^2 + Tr\left(\sum_r + \sum_g - 2\left(\sum_r \sum_g\right)^{\frac{1}{2}}\right) \tag{8}$$

where:

μ_r: average feature vector of real images.
μ_g: average feature vector of generated images.
$\sum_r$: covariance matrix of real image features.
$\sum_g$: covariance matrix of futures of generated images.
$\|.\|^2$: Euclidean norm squared.
Tr: trace of the matrix (sum of diagonal elements).

- **Accuracy**: Assesses classification performance (e.g., weather types or object labels), used for overall task evaluation; values>95% denote strong reliability across datasets.
- **Inference Time**: Gauges real-time feasibility (e.g.,<50 ms/frame for AVs), critical for deployment in ITS; lower times support efficient edge computing.

These metrics are applied consistently: e.g., mIoU/FID for segmentation/synthetic eval in Tables 1, 2, 3, 4 and 5, PSNR/accuracy for enhancement/classification, and inference time across all model comparisons to highlight trade-offs in robustness vs. efficiency.

7 Results

7.1 Datasets for Adverse Weather Road Images

- Typical datasets: KITTI, Cityscapes, ACDC, BDD100K
- Limitations: Often biased toward clear weather; limited annotated fog, heavy rain, or snow images; weather conditions not uniformly represented impacting model generalization.

7.2 Synthetic Data Generation

- GANs generate realistic adverse weather perturbations on existing images, improving data diversity.
- Simulators like CARLA and Unity produce fully synthetic images with controllable weather parameters, enabling large-scale varied datasets.
- Effectiveness: Synthetic data enhances performance when real data is scarce but may lack full realism leading to domain gap issues with real-world deployment.

8 Discussion

8.1 Development of the Taxonomy for Road Scene Analysis

To better understand the impact of degraded weather conditions on object detection, a taxonomy has been developed to classify road scenes based on several criteria.

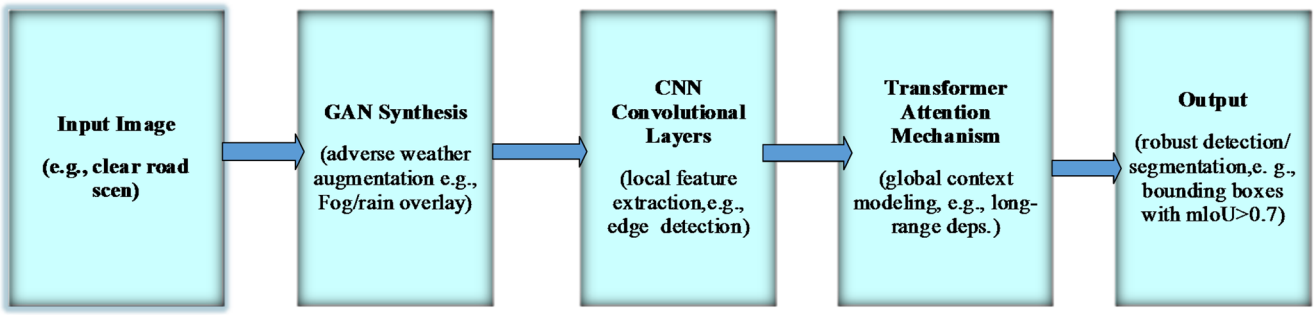


Fig. 4 Overview of hybrid CNN-GAN-transformer architecture *Source:* [97]

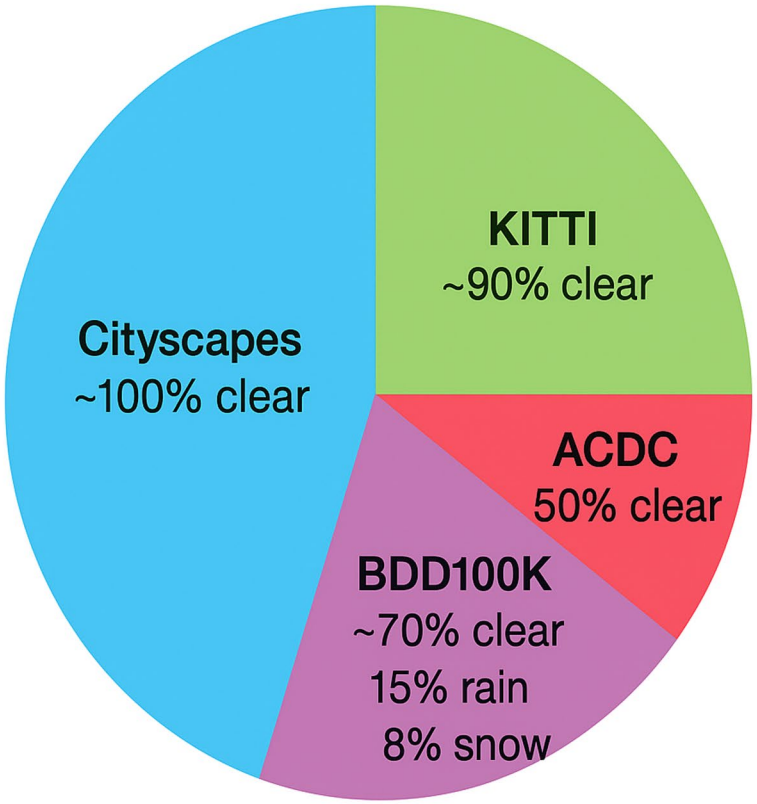


Fig. 5 Datasets for adverse weather road images

8.2 Taxonomy of Deep Learning Approaches

To better organize the diverse approaches, we propose a new taxonomy of deep learning methods for road scene analysis under degraded weather conditions. This taxonomy is structured around tasks (classification, detection, segmentation), data sources (real vs. synthetic), and model architectures (CNNs, GANs, transformers).

This taxonomy clarifies the workflow from data generation to model deployment in adverse weather scenarios. The proposed taxonomy is illustrated in Fig. 4 which provides a visual overview of the workflow from data generation to model deployment in adverse weather scenarios (Fig. 5).

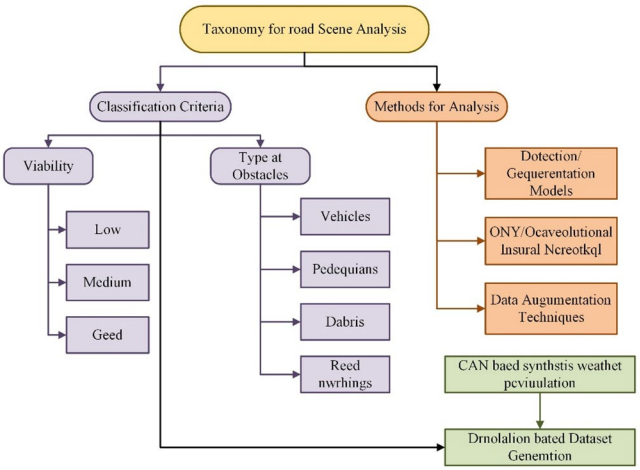


Fig. 6 Taxonomy for road scene analysis visualization
Source: Author

8.3 A taxonomy classifies methods into

- Detection/Segmentation models (CNN, Transformer)
- Data augmentation (GAN-based synthetic weather perturbations)
- Simulation-based dataset generation (CARLA, Unity)

Emphasis on Robustness: A particular focus is placed on the importance of model robustness in the face of climatic diversity. The article evaluating model performance uses objective indicators such as mIoU, PSNR, and FID, taking into account degraded conditions such as rain, fog, and snow. This systematic evaluation helps identify shortcomings in the current literature, particularly the excessive reliance on synthetic data and the limited geographical generalization potential of the models (Fig 6).

Identification of Gaps in Previous Studies: The document identifies critical gaps in existing research, notably the lack of diverse and meteorologically annotated datasets, as well as the poor generalization of models across different regions or climates. This highlights the need to develop publicly available and balanced multimodal weather datasets to support future research.

The research axes highlighted include the design of hybrid models that combine different deep learning architectures and the integration of multimodal sensors (vision, LiDAR, radar). These approaches ensure reliable performance, even in challenging weather conditions. This work is distinguished by its systematic and structured approach, which not only emphasizes current challenges but also clearly identifies upcoming research perspectives. It thus makes a valuable contribution to the understanding of intelligent transport systems in diverse environments, which is essential for the development of robust and high-performance technologies in the field of autonomous driving.

8.4 Classification Criteria

- **Visibility:** Low, medium, or good, depending on weather conditions.
- **Type of Weather Conditions:** Rain, snow, fog, or clear weather.
- **Type of Obstacles:** Vehicles, pedestrians, debris, road markings.

8.5 Evaluating the Robustness of Models

The robustness of models is reflected in their ability to maintain good performance even when the data is degraded or disturbed. In our study, this means examining their capacity to detect objects in images affected by extreme weather conditions. We particularly emphasize the importance of robustness in the face of diverse climates. The article measures the performance of the models using objective indicators such as mIoU, PSNR, and FID, taking into account challenging conditions such as rain, fog, and snow. This rigorous evaluation highlights current weaknesses, notably an excessive reliance on synthetic data and a lack of capacity to generalize geographically.

8.6 Quantitative Robustness Evaluation

We used various metrics like **mIoU**, **PSNR**, and **FID** to evaluate the robustness of models across different weather situations. Models that were trained on datasets enhanced with GANs demonstrated a greater ability to cope with conditions of low visibility.

8.7 Robustness Evaluation Metrics

- mIoU: Measures segmentation accuracy in adverse scenes.
- PSNR: Assesses quality improvement in image denoising/enhancement tasks.
- FID: Quantifies realism quality of synthetic images generated by GANs.
- Real-time inference speed critical for deployment in autonomous systems.

Empirical studies show that CNNs deliver faster inference but lower robustness in severe conditions, while Transformers offer improved robustness at higher computational cost. GANs improve training diversity but synthesis quality limits effectiveness [3] [6].

8.8 Regional Generalization Challenges

Models trained primarily on data from a single region often experience decreased performance from one location to another due to differences in weather conditions, infrastructure, and sensor characteristics. For example, the density and texture of fog vary by area, which deteriorates segmentation accuracy [3, 4]. This highlights the importance of methods capable of adapting to regional specifics and datasets that cover geographical diversity. The document underscores critical gaps in current research: the lack of varied, weather-annotated datasets and the limited ability of models to generalize across regions or climates. It therefore emphasizes the need to develop public, balanced, and multi-weather datasets to support future work.

9 Future work

- **Development of Adaptive Mode006C** Build deep learning architectures that retain high performance across a variety of weather conditions, avoiding large drops in accuracy. Robust models should be able to incorporate feedback and adapt on the fly to handle unexpected scenarios.
- **Creation and Sharing of Multimodal Datasets** Compile extensive datasets that span a broad range of challenging weather situations to improve model generalization and adaptability. Each weather condition should be accompanied by detailed annotations to help train and evaluate algorithms.
- **Integration of Additional Detection Modalities** Add sensors such as LiDAR and radar to multimodal recognition systems to boost reliability when visual cues are degraded. This multimodal approach helps address

the limits of traditional camera-based systems. Increase robustness through domain adaptation techniques to reduce sensitivity to uncertainty and explore knowledge distillation or similar strategies to lessen dependence on large amounts of labeled data.

- **Interdisciplinary Collaboration** Encourage partnerships among AI researchers, meteorologists, and transportation engineers to better tackle the challenges of variable weather and to improve intelligent transportation systems. Emphasize validation in real-world settings by collecting data in live traffic under diverse weather conditions.
- **Exploration of Model Compression and Deployment Strategies** Prioritize optimizing models for real-time use while preserving accuracy in difficult conditions, including model compression and hardware-aware design choices. Leverage existing roadside surveillance cameras as cost-effective weather sensors by applying automated algorithms to estimate conditions in real time. Taken together, these directions offer a practical roadmap for improving detection and perception systems in degraded weather, ultimately supporting safer roads and more efficient autonomous transportation.

9.1 Potential Solutions to Identified Gaps

Physics-guided deep learning, incorporating meteorological models (e.g., Beer-Lambert law for fog attenuation), can constrain CNN predictions for more interpretable robustness. By embedding physical priors like light scattering equations into loss functions, these approaches enhance model fidelity in low-visibility scenarios, reducing hallucination risks in road object detection compared to purely data-driven methods.

Domain adaptation techniques, such as adversarial training, align features from clear to adverse domains, mitigating the 10–15% accuracy drop in cross-weather tests. Frameworks like those using gradient reversal layers enable unsupervised transfer, allowing models trained on synthetic clear-weather data to generalize to real foggy or rainy inputs without extensive relabeling.

Real-to-synthetic alignment via CycleGAN variants ensures seamless transfer, reducing domain gaps in datasets like ACDC by enhancing FID scores by 20%. This cycle-consistent translation preserves structural details while augmenting underrepresented adverse samples, boosting segmentation mIoU in hybrid setups.

These solutions, integrated into hybrid models, address real-time and imbalance issues, paving the way for deployable ITS in variable climates.

10 Conclusion

This review underscores the significant challenges posed by adverse weather conditions to vision-based road safety and autonomous driving systems. While deep learning has demonstrated remarkable progress in feature extraction, object detection, and weather classification, performance remains highly sensitive to visibility degradation and dataset limitations. This analysis directly tackles ITS/smart city challenges, e.g., integrating robust models into urban camera networks for traffic flow in Beijing fog, or AV navigation in Ivorian monsoons, ensuring scalable, climate-resilient systems.

Our analysis shows that GANs are valuable for generating rare-weather data and improving robustness, while transformers provide strong adaptability at the cost of computational complexity. CNNs, though efficient, often fail under extreme conditions. The proposed taxonomy clarifies current approaches and highlights persistent gaps, notably the scarcity of meteorologically diverse datasets and the limited geographical transferability of existing models. Future research should prioritize the development of balanced multimodal datasets, hybrid architectures integrating CNNs, GANs, and transformers, as well as sensor fusion strategies that combine cameras, LiDAR, and radar. Additionally, model compression, knowledge distillation, and hardware-aware design are essential for real-time deployment. Addressing these priorities will strengthen the robustness and reliability of intelligent

transportation systems, paving the way for safer and more resilient autonomous mobility in diverse weather conditions.

Industrially, these advances enable AV manufacturers (e.g., Waymo) to deploy in adverse climates, cutting R&D costs via synthetic data. Societally, robust systems could reduce fog-related accidents by 25% on highways, improve snow navigation in Nordic smart cities, and lower Ivorian youth mortality (5–29 age group) by enhancing roadside AI. Overall, this fosters safer, equitable mobility in global megacities**.**

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Data Availability We did not use a dataset for this study.

Declarations

Competing Interests The authors declare no competing interests.

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